Farhan Sadik

24101406, Sec-19

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## Que 1:

The Monetisation of Health Data case study tells the account of the mass breach of a significant volume of personal information when the health and fitness data of nearly 61 million users of a health-tracking company started to be publicly available. This case shows how massive commercial exploiting of health data may have negative impacts on individuals, society, and sustainability in the long-term (Stahl et al., 2023).

Health data are one of the most personal details that may be sensitive in the given society. Their exposure may result in employer and insurance discrimination, loss of privacy, psychological upheavals and will result to loss of user confidence. The book describes the fact that there was no noticeable fact that users reasonably thought that the information about their health activity would be shared or monetised in addition to providing basic services, which shows the failure in terms of autonomy and informed consent (Stahl et al., 2023). In the long run, these events can weaken the desire of the population to establish digital health technologies, which hamper their potential to benefit the health research of the population.

Similar cases were observed in the actual implementation of the digital health system. As an illustration, in the case of the Royal Free NHS Trust which has distributed 1.6 million patient records to Google DeepMind without adequate transparency, nor any consent, the Information Commissioner of the UK judged that the data monetization of patient information in the healthcare context breached the privacy expectations of patients, even though in a trust care institution (Information Commissioner’s Office [ICO], 2017). These instances bring out repeating governance breakdowns in massive health data utilization.

In the long term, there is the normalization of surveillance capitalism since people will be used mainly as source of data, not as holders of rights. Such systems, as Zuboff (2019) claims, will transfer authority to corporations and undermine personal control over personal information. This negates the transparency of democracy and social confidence in online systems. There are heavy power resources and carbon emissions because of continuous data collection, cloud storage and large scale data analytics. One of the studies demonstrates that supporting data-intensive AI systems has quantifiable environmental costs (Strubell et al., 2019). Secondly, health-tracking gadgets promote the electronic waste as hardware updates are easily brought.

**Que 2:**

In general, the system outlined cannot be sustainable as it is. The monetisation of health data will result in greater disadvantages to the society compared to advantage unless there is solid protection of the data, open governance, and taking into account environmental impacts.

The system outlined in the case study contains a number of ethical breaches in the areas of transparency, accountability, and possible prejudice. These issues are closely related although the major concern is that of privacy.

To begin with, there is a great lack of transparency. The means of storing, protecting, and even commercialising the health information of users was not explained to the user clearly. The book reports that data leakage has shown that the records of user activity were publicly accessible and not known that their information was being announced, which was against the assumptions of confidentiality (Stahl et al., 2023). Ethical AIs systems demand that people comprehend the manner in which choices on their data are undertaken, and the purpose of the decision making.

Second, there is poor accountability. With the breach, effective responsibility-taking by the company in form of proper user notification, compensation, or structural reform is probable, which was lacking after the breach. Accountability is vital in AI ethics where organisations should be held accountable in case of harm by the systems (Wachter and Mittelstadt, 2019).

Third, the system is not discriminatory, but indirect bias might be present upon monetisation of the health data. Fitbit data tends to overrepresent wealthy and urban folk. Discriminated groups can be prejudiced, or even discriminated, against in case such data is utilized by insurers or employers later on. These types of risks have also been reported in health algorithms that reproduce racially and socioeconomically biased situations (Obermeyer et al., 2019).

Lastly, it can be stated that the case features the attributes of the surveillance capitalism in which behavioural information is harvested outside the reasonable expectations of users to generate profits (Zuboff, 2019). This compromises moral values of equity, independence and human dignity. To conclude, the system is unethical because it lacks transparency, accountability, and guard against biased misuse of the sensitive health information in a downstream way.

**Que 3:**

Implementation of the health data monetisation system as outlined in the case study would have high legal consequences especially in regards to data protection and privacy laws. Health information is categorised under the legal term of sensitive personal data and thus safeguarded on the ground of stricter protection compared to normal personal information (Stahl et al., 2023).

The processing of health data under EU general data protection regulation (GDPR) must have explicit user consent, ensure that the purpose of the process is limited, and there should be suitable technical and organisational control over the process. The fact that millions of users have their health records on part of them has been publicly disclosed would probably amount to significant violation of GDPR, which may lead not only to administrative fines finely up to 4 percent of annual global turnover, but also cause regulatory inquiries and the suspension of data processing operations (European Union, 2016). The same happened in the case with Google DeepMind-NHS royal free where because patient information was not available with enough transparency and consent, the information was being disseminated against the law ( Information Commissioners Office [ICO], 2017).

Outside of regulatory fines, organisations can be sued in civil courts on the account of negligence when poor security measures result in preventable damage, e.g. identity theft, discrimination, or even mental distress, in cases where post-breach accountability is low.

Before rolling such systems, engineers are instrumental in making sure that they are safe, fair and in line with the law. They need to adopt privacy-by-design and security-by-design with encryptions, access controls, auditing, and secure cloud-setups, and be transparent with a clear documentation of data collection, data flows, and third party access.

When working with sensitive health information, engineers are also to carry out Data Protection Impact Assessments (DPIAs) and AI Impact Assessments to detect risks and prevent them at an early stage (Floridi et al., 2018). They should also take into account possible misuse of data by insurers or employers downstream and develop mechanisms that can reduce the effects of discrimination with the help of legal and ethical professionals.

To sum up, the lack of these legal and professional duties leads organisations to severe legal outcomes, and to the trustful and non-illegal AI systems, responsible engineering activities are necessary.

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